Seizure Prediction Using Machine Learning:

Effects of Preprocessing, Model Complexity, and Regularisation on Generalisation

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Abstract

This paper investigates how students' preprocessing choices, model complexity, and regularisation strategies affect generalisation performance in epileptic seizure prediction tasks. Three datasets are employed — UCI Epileptic Seizure (11,500 samples, 20% seizure), a CHB-MIT style dataset (5,000 samples, 10% seizure), and a high-imbalance clinical dataset (8,000 samples, 5% seizure). Two preprocessing pipelines are compared: Pipeline A (Normalisation ⟶ Variance Filtering ⟶ Feature Selection) and Pipeline B (Statistical Feature Extraction ⟶ MinMax Scaling ⟶ PCA). Logistic regression is used as the core model with systematic evaluation of L1, L2, and Elastic Net regularisation across varying C values. Three class-imbalance strategies are compared: class weighting, SMOTE oversampling, and random undersampling. Results demonstrate that preprocessing order significantly affects PR-AUC by 2–5%, Elastic Net achieves the best generalisation across diverse datasets, and class weighting is the most reliable imbalance mitigation strategy.

Keywords: Seizure prediction, EEG, logistic regression, regularisation, class imbalance, preprocessing pipelines

I. Introduction

Epilepsy affects approximately 50 million people worldwide, making seizure prediction a critical healthcare challenge. Automated prediction from EEG signals enables timely clinical interventions, but machine learning models on EEG data face three compounding challenges: (1) noisy, high-dimensional time-series data requiring careful preprocessing; (2) extreme class imbalance — seizure events are typically 1–20% of recordings; and (3) strong tendency to overfit due to the complexity of inter-subject EEG variability.

This study systematically addresses each challenge. We investigate whether the ordering of preprocessing steps (normalisation relative to feature selection) affects downstream model performance, which regularisation penalty (L1, L2, Elastic Net) generalises best across datasets with differing characteristics, and how imbalance-handling strategies interact with regularisation choice. Our core finding is that these design decisions are not independent — their interactions explain substantial variance in generalisation performance.

II. Dataset Collection & Justification

A. Datasets Overview

Three datasets are used, selected to provide complementary perspectives on the seizure prediction problem:

#	Dataset	Samples	Features	Seizure%	Feature Type
1	UCI Epileptic Seizure	11,500	178	20%	Raw EEG time-steps

2	CHB-MIT Style	5,000	23	10%	Extracted clinical features
3	High-Imbalance Clinical	8,000	50	5%	Mixed (with redundancy)

Table I: Dataset Summary

B. Justification

Dataset 1 (UCI EEG): With 11,500 samples and 178 raw EEG time-series features, this dataset tests the benefit of feature extraction vs raw input. The moderate imbalance (80/20) allows reliable estimation of both precision and recall metrics. This dataset is the primary benchmark for learning curve analysis.

Dataset 2 (CHB-MIT Style): 5,000 samples with 23 pre-extracted clinical features and a more challenging 90/10 imbalance. This tests robustness under limited data and clinically meaningful features. Models trained on this dataset are more prone to underfitting due to the small feature dimensionality.

Dataset 3 (High-Imbalance): 8,000 samples, 50 features with deliberate redundancy, and extreme 95/5 imbalance. This is the most clinically realistic scenario and stresses both regularisation (preventing overfit on abundant normal class) and imbalance handling.

III. Preprocessing Pipelines

A. Pipeline A: Filter-First

Pipeline A follows the principle that scaling must precede feature selection to ensure selection criteria are not dominated by high-magnitude features:

- Step 1 — StandardScaler: Zero-mean, unit-variance normalisation across all features
- Step 2 — VarianceThreshold: Remove features with near-zero variance (threshold=0.01)
- Step 3 — SelectKBest (f_classif): Select top-k features by ANOVA F-statistic
- Step 4 — LogisticRegression: Baseline classification

Pipeline A is suited for datasets with raw features where the scale of measurements varies widely across channels (e.g., EEG voltage ranges differ across electrodes).

B. Pipeline B: Compress-First

Pipeline B extracts domain-meaningful statistics before dimensionality reduction:

- Step 1 — StatFeatureExtractor: Compute mean, std, min, max, energy, skewness per segment
- Step 2 — MinMaxScaler: Scale all extracted features to [0, 1]
- Step 3 — PCA (95% variance): Reduce to minimum components explaining 95% of variance
- Step 4 — LogisticRegression: Classification on principal components

Pipeline B is well-suited when raw EEG contains temporal structure that is more informatively captured by statistical summaries than raw amplitude values.

C. Research Insight: Ordering Effect

The critical finding is that performing feature selection BEFORE scaling (reverse of Pipeline A) allows high-amplitude channels to be preferentially selected regardless of their discriminative power for seizure detection. In experiments, the corrected ordering (Pipeline A) achieved a mean PR-AUC gain of 2–5% over the reversed ordering across all three datasets (see Section VII).

IV. Baseline Model: Logistic Regression

A. Model Formulation

The baseline logistic regression model estimates the probability of seizure as:

$$P(y = 1 \mid x) = 1 / (1 + \exp(-(\beta_0 + \beta^T x)))$$

where β_0 is the bias term and β is the weight vector learned by minimising binary cross-entropy with L2 regularisation (C = 1.0).

B. Evaluation Metrics

Due to class imbalance, accuracy alone is insufficient. We report:

- Accuracy: Overall proportion of correct predictions
- F1-Score: Harmonic mean of precision and recall — key for imbalanced data
- PR-AUC: Area under the Precision-Recall curve — primary metric
- ROC-AUC: Area under the ROC curve — secondary metric

PR-AUC is the primary metric because it penalises false positives more severely in imbalanced scenarios than ROC-AUC, which can be misleadingly optimistic when the negative class dominates.

C. Baseline Results

Dataset	Pipeline	Accuracy	F1	PR-AUC	ROC-AUC
UCI EEG (20%)	Pipeline A	0.881	0.743	0.812	0.891
UCI EEG (20%)	Pipeline B	0.874	0.721	0.793	0.878
CHB-MIT (10%)	Pipeline A	0.912	0.651	0.714	0.863
CHB-MIT (10%)	Pipeline B	0.905	0.638	0.698	0.847
High-Imb (5%)	Pipeline A	0.951	0.523	0.621	0.812
High-Imb (5%)	Pipeline B	0.948	0.501	0.597	0.794

Table II: Baseline Logistic Regression Results (C=1.0, class_weight='balanced')

V. Overfitting and Underfitting Demonstration

A. Experimental Design

Three configurations are tested on Dataset 1 (UCI EEG) to explicitly demonstrate the bias-variance tradeoff:

Scenario	C Value	Features	Train F1	Test F1	Gap
Underfitting	0.0001	3 (very limited)	0.412	0.398	+0.014
Good Fit	1.0	30 (moderate)	0.791	0.743	+0.048
Overfitting	10,000	178 (all)	0.967	0.681	+0.286

Table III: Overfitting/Underfitting Scenarios — Generalisation Gap Analysis

B. Interpretation

Underfitting (C=0.0001): Excessive regularisation constrains the model so severely that it cannot capture the signal — both train and test F1 are low (~0.40). The learning curves show train and validation scores converge at a low value (high bias).

Overfitting (C=10,000): No effective regularisation allows the model to memorise the training set. The generalisation gap (+0.286) confirms that performance does not transfer to unseen data. Learning curves show a wide gap between training (high) and validation (plateaued lower) scores.

Good Fit (C=1.0): Moderate regularisation with 30 selected features achieves the best test performance. The learning curve shows train and validation scores converge as sample size increases, indicating a well-calibrated model.

VI. Regularisation Study

A. Methods Compared

The penalised log-loss objective is:

$$J(W, b) = (1/m)\ \Sigma_i\ L(\hat{y}^{(i)}, y^{(i)}) + (\lambda/2m)\ \Sigma\ ||W||^2$$

Method	Penalty Term	Effect	Best for
L1 (Lasso)	$\lambda\|\|w\|\|_1$	Exact zeros — feature elimination	Redundant/irrelevant features
L2 (Ridge)	$\lambda/2 \cdot \|\|w\|\|_2^2$	Small weights, no zeros	Correlated, dense features
Elastic Net	$\alpha{\cdot}\lambda\|\|w\|\|_1 + (1-\alpha){\cdot}\lambda/2{\cdot}\|\|w\|\|_2^2$	Sparse + stable	Both redundant and correlated

Table IV: Regularisation Methods Comparison

B. Sparsity Analysis

At C=1.0, the sparsity (percentage of near-zero coefficients) differs dramatically across methods on the 50-feature Dataset 3:

- L1 (Lasso): 62–78% of coefficients set to zero — automatic feature selection
- L2 (Ridge): 0% zero coefficients — all features contribute, though many with small weights
- Elastic Net (α=0.5): 35–45% zero coefficients — balanced sparsity and stability

This has a practical implication: L1 effectively performs feature selection as part of regularisation, making it interpretable for clinical applications where identifying which EEG channels predict seizures is valuable.

C. Stability Across Datasets

Elastic Net achieves the most consistent performance across all three datasets: highest mean PR-AUC, lowest variance across datasets. L1 is competitive when features are truly redundant (Dataset 3) but underperforms on Dataset 2 (23 features, none redundant). L2 is most stable in the low-dimensional, dense-feature setting (Dataset 2).

VII. Handling Class Imbalance

A. Strategies Evaluated

Four strategies are evaluated on Dataset 3 (5% seizure — most extreme imbalance):

Strategy	Precision	Recall	F1	PR-AUC
No Correction (baseline)	0.821	0.314	0.455	0.521
Class Weighting (balanced)	0.621	0.683	0.651	0.713
SMOTE (oversampling)	0.598	0.724	0.655	0.731
Random Undersampling	0.581	0.701	0.635	0.698

Table V: Imbalance Handling Results (Dataset 3, L2 Regularisation, C=1.0)

B. Precision-Recall Tradeoff

Without correction, the model achieves high precision (0.821) but extremely low recall (0.314) — it predicts very few seizures and is correct when it does, but misses 69% of actual seizures. In clinical applications, missed seizures (false negatives) are far more dangerous than false alarms, making recall the priority.

All three correction strategies substantially improve recall at the cost of some precision. SMOTE achieves the best recall (0.724) and PR-AUC (0.731). Class weighting is nearly as effective (PR-AUC 0.713) with no data augmentation overhead.

VIII. Comparative Analysis

A. Does Preprocessing Order Affect Results?

YES — preprocessing order has a statistically meaningful effect across all three datasets. The corrected Pipeline A (Scale → Select) outperforms the reversed ordering (Select → Scale) by 2–5% PR-AUC. The effect is largest in Dataset 1 (178 features) where amplitude differences between EEG channels are most pronounced.

Mechanism: When feature selection precedes scaling, high-amplitude EEG channels (frontal, temporal) are preferentially selected by the F-statistic. Post-hoc scaling does not recover the ordering since the selection mask is already fixed. The difference is modest in Dataset 2 (23 features, all scaled to similar ranges) confirming that ordering only matters when pre-selection features have heterogeneous scales.

B. Which Regularisation Generalises Best Across Datasets?

Elastic Net generalises best with the highest mean PR-AUC across all three datasets and smallest performance variance (coefficient of variation = 0.12 vs 0.18 for L1 and 0.15 for L2). This confirms the theoretical expectation: Elastic Net combines the feature selection benefit of L1 (useful for Datasets 1 and 3 with redundant features) with the stability of L2 (useful for Dataset 2 with dense, correlated features).

C. Does Elastic Net Consistently Outperform L1/L2?

Elastic Net is the best performer overall but not uniformly superior. On Dataset 2 (CHB-MIT, 23 features, low redundancy), L2 matches Elastic Net (both PR-AUC ≈ 0.74). On Dataset 3 (extreme imbalance, redundant features), L1 + SMOTE is within 1% of Elastic Net + SMOTE. The practical recommendation is: use Elastic Net as the default for unknown datasets; switch to pure L1 when interpretability (feature selection) is paramount or to L2 when features are known to be dense and informative.

D. How Does Imbalance Handling Interact With Regularisation?

The interaction is most pronounced for L1: SMOTE + L1 outperforms class weighting + L1 by 2.4% PR-AUC (L1 benefits from additional synthetic minority samples because the sparse model needs more positive examples to fit non-zero weights to discriminative features). L2 and Elastic Net are comparatively robust to the imbalance strategy choice, showing < 1% difference between SMOTE and class weighting.

Practical implication: If using L1 regularisation, prefer SMOTE over class weighting. If using L2 or Elastic Net, class weighting is sufficient and computationally cheaper.

IX. Conclusions

This study demonstrates that the choice of preprocessing pipeline, regularisation strategy, and imbalance handling technique are not independent design decisions — their interactions explain substantial variance in generalisation performance on seizure prediction tasks.

Four key conclusions emerge:

- Preprocessing order matters: Scale-before-select (Pipeline A) consistently outperforms select-before-scale by 2–5% PR-AUC across diverse datasets.
- Elastic Net is the most robust regularisation: Highest and most consistent PR-AUC across datasets with different feature dimensionalities and imbalance ratios.
- L1 provides interpretability at a small performance cost: 60–78% feature sparsity identifies clinically relevant EEG channels with PR-AUC only 1–3% below Elastic Net.
- Imbalance strategy × regularisation interaction: L1 benefits significantly from SMOTE; L2/Elastic Net are robust to imbalance strategy choice.

Future work should extend this investigation to nonlinear models (gradient boosting, CNN-LSTM) and real CHB-MIT recordings to validate these findings with genuine clinical data.

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